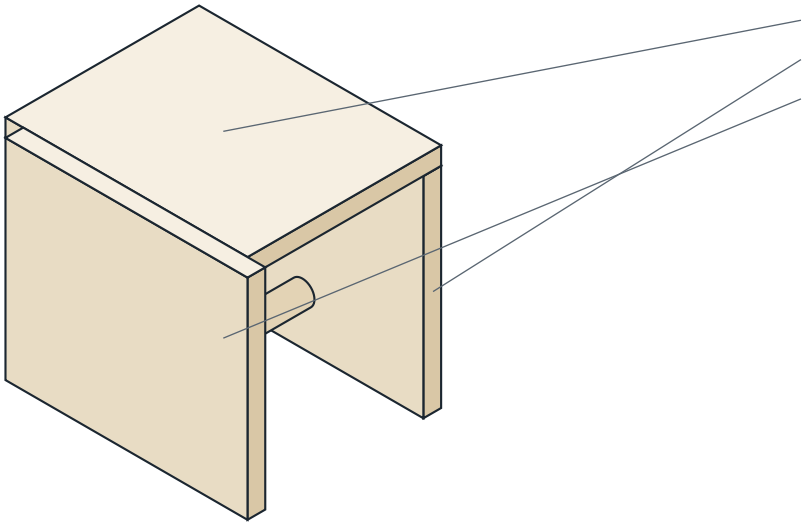
Case study 5: Step stool

National 5 · Practical Woodworking · 10 marks · open book

The product. A small step stool for a kitchen has two legs, a top step and a turned rail between the legs for strength. Use the drawing to answer the questions. Answer in sentences for describe and explain questions; you can add labelled sketches.



Main sizes (mm)

Height 282
Width 300
Depth 240
Rail Ø30, turned between centres

1 *The turned rail is made on the woodturning lathe between centres.*

(a) Describe how to prepare the square blank before mounting it on the lathe. **(2 marks)**

(b) Describe two safety precautions when using the lathe. **(2 marks)**

2 *The rail must be 30 mm in diameter.*

Name the measuring tool used to check the diameter while turning. **(1 mark)**

3 *The top step is made from beech.*

Explain why beech is suitable for the step. **(2 marks)**

4 *The legs are joined to the top with a flat-frame or carcase joint.*

Name a suitable joint and give one reason for your choice. **(2 marks)**

5 State one item of PPE that should not be worn when using the lathe, and why. **(1 mark)**

Total: ___ / 10